

Change log

Centrality analysis on Rio Grande do Sul

Revision	Date	Author	Description
1.0	2025.04.01 12:41	BSH	Created
1.0	2025.04.01 23:56	RRS	Correction

Summary

1. **Fixed** bugs in Rmarkdown v0 due to wrong names of the variables of the objects ("st union" changed to "geom"). These changes were 231c251, 265c286, 267c288 270c291 and 326c347.
2. **Fixed** bug in Rmarkdown v0 due to wrong parameters in code chunks. This change was 235c255
3. **Fixed** bug in Rmarkdown v0 due to wrong figures directory 125c145,146, 155c176, 195c216, 224c245, 255c276, 301c322 and 309c330.
4. **Fixed** minor errors. These changes were 245c266, 248c269, 261c282, 294c315, 324c345, 237c257, 258 and 331c352.
5. **Added** code in Rmarkdown v0 chunk A to load the temp data. This change was 94a95,114.
6. **Added** code in Rmarkdown v0 to calculate results instead of typing. This change was 259c280 and 354c375.

The command line in linux (ubuntu) to obtain the difference between the revisions was:

```
colordiff --suppress-common-lines quarto\_script\_rs\_paper\_v0.Rmd
quarto\_script\_rs\_paper\_v1\_Rmd
```

Or

```
diff quarto\_script\_rs\_paper\_v0.Rmd quarto\_script\_rs\_paper\_v1\_Rmd
```

We recommend to use the command for an accurate version of the full detailed version history.

Full detailed version history

1. Renaming variables of objects

231c251

< The quantitative analysis of road connectivity before and after the flood revealed an overall connectivity decrease in the analysed `r
units::drop_units(round(sum(st_length(centrality_pre\$geom))/1000))` km of urban roads and surrounding `units::drop_units(round(st_area(flood_extent\$**st_union**)/10000))` ha of flood extent.

> The quantitative analysis of road connectivity before and after the flood revealed an overall connectivity decrease in the analysed `r
units::drop_units(round(sum(st_length(centrality_pre\$geom))/1000))` km of urban roads and surrounding `units::drop_units(round(st_area(flood_extent\$**geom**)/10000))` ha of flood extent.

265c286

< However, the centrality of this avenue experienced only a a `r
round((filter(local_connectivity_post, fid == 95421)\$centrality - filter(local_connectivity_pre, **fid** == 95421)\$centrality)/filter(local_connectivity_pre, **fid** == 95421)\$centrality*100,2)`% reduction.

> However, the centrality of this avenue experienced only a a `r
round((filter(local_connectivity_post, id == 95421)\$centrality - filter(local_connectivity_pre, **id** == 95421)\$centrality)/filter(local_connectivity_pre, **id** == 95421)\$centrality*100,2)`% reduction.

267c288

< While the flood reduced the connectivity of the central road Avenida João Pereira de Vargas for Sapucaia do Sul by a `r round((filter(local\_connectivity\_post, fid == 144543)\$centrality - filter(local\_connectivity\_pre, **fid** == 144543)\$centrality)/filter(local\_connectivity\_pre, **fid** == 144543)\$centrality\*100,2)`%, the central road Rua Florianópolis in Canoas was covered by the flood shown in the Fig. 6D, resulting in a complete loss of connectivity (-100%).

> While the flood reduced the connectivity of the central road Avenida João Pereira de Vargas for Sapucaia do Sul by a `r round((filter(local\_connectivity\_post, id == 144543)\$centrality - filter(local\_connectivity\_pre, **id** == 144543)\$centrality)/filter(local\_connectivity\_pre, **id** == 144543)\$centrality\*100,2)`%, the central road Rua Florianópolis in Canoas was covered by the flood shown in the Fig. 6D, resulting in a complete loss of connectivity (-100%).

270c291

< In a comparative analysis of the Nova Santa Rita and Canoas cases, we observed how the `r units::drop\_units(round(st\_area(st\_intersection(flood\_extent, filter(connectivity\_municipality\_lc\_rq, municipality=="Nova Santa Rita"))\$**st\_union**)/1000000))` km² (`r round(units::drop_units(st_area(st_intersection(flood_extent,

```
filter(connectivity_municipality_lc_rq,municipality=="Nova Santa
Rita"))$st_union)/1000000)/units::drop_units(st_area(filter(connectivity_municipality_lc_rq,m
unicipality=="Nova Santa Rita")$geom)/1000000) * 100,2)`%) flooded extension in Nova
Santa caused a loss of intracity connectivity of `r round((sum(filter(local_connectivity_post,
shapename == "Nova Santa Rita")$centrality) - sum(filter(local_connectivity_pre,
shapename == "Nova Santa Rita")$centrality)) / sum(filter(local_connectivity_pre,
shapename == "Nova Santa Rita")$centrality) * 100,2)`% , while the `r
units::drop_units(round(st_area(st_intersection(flood_extent,
filter(connectivity_municipality_lc_rq,municipality=="Canoas"))$st_union)/1000000))` km²
(`round(units::drop_units(st_area(st_intersection(flood_extent,
filter(connectivity_municipality_lc_rq,municipality=="Canoas"))$st_union)/1000000)/units::dro
p_units(st_area(filter(connectivity_municipality_lc_rq,municipality=="Canoas")$geom)/10000
00) * 100,2)`%) of flood in Canoas caused a loss of intracity connectivity of `r
round((sum(filter(local_connectivity_post, shapename == "Canoas")$centrality) -
sum(filter(local_connectivity_pre, shapename == "Canoas")$centrality)) /
sum(filter(local_connectivity_pre, shapename == "Canoas")$centrality) * 100,2)`%.
```

```
> In a comparative analysis of the Nova Santa Rita and Canoas cases, we observed how the
`r units::drop_units(round(st_area(st_intersection(flood_extent,
filter(connectivity_municipality_lc_rq,municipality=="Nova Santa Rita"))$geom)/1000000))`
km² (`r round(units::drop_units(st_area(st_intersection(flood_extent,
filter(connectivity_municipality_lc_rq,municipality=="Nova Santa
Rita"))$geom)/1000000)/units::drop_units(st_area(filter(connectivity_municipality_lc_rq,muni
cipality=="Nova Santa Rita")$geom)/1000000) * 100)`%) flooded extension in Nova Santa
caused a loss of intracity connectivity of `r round((sum(filter(local_connectivity_post,
shapename == "Nova Santa Rita")$centrality) - sum(filter(local_connectivity_pre,
shapename == "Nova Santa Rita")$centrality)) / sum(filter(local_connectivity_pre,
shapename == "Nova Santa Rita")$centrality) * 100,2)`% , while the `r
units::drop_units(round(st_area(st_intersection(flood_extent,
filter(connectivity_municipality_lc_rq,municipality=="Canoas"))$geom)/1000000))` km²
(`round(units::drop_units(st_area(st_intersection(flood_extent,
filter(connectivity_municipality_lc_rq,municipality=="Canoas"))$geom)/1000000)/units::drop_
units(st_area(filter(connectivity_municipality_lc_rq,municipality=="Canoas")$geom)/1000000
) * 100,2)`%) of flood in Canoas caused a loss of intracity connectivity of `r
round((sum(filter(local_connectivity_post, shapename == "Canoas")$centrality) -
sum(filter(local_connectivity_pre, shapename == "Canoas")$centrality)) /
sum(filter(local_connectivity_pre, shapename == "Canoas")$centrality) * 100,2)`%.
```

326c347

```
< Considering a 10-minute isochrone, the `r
stringr::str_to_title(arrange(mutate(isochrone_ghs_aoi,area =
st_area(isochrone_ghs_aoi$geometry)),desc(area))$ds_cnes[1])` with the highest lack of
redundancy reached an area that was `r round(arrange(mutate(isochrone_ghs_aoi,area =
st_area(isochrone_ghs_aoi$geometry)),desc(area))$area[1]/arrange(mutate(isochrone_ghs_
aoi,area = st_area(isochrone_ghs_aoi$geometry)),area)$are[1],2)` times larger than that of
the hospital `r arrange(mutate(isochrone_ghs_aoi,area =
st_area(isochrone_ghs_aoi$geometry)),area)$ds_cnes[1]`.
```

```
> Considering a 10-minute isochrone, the `r
string::str_to_title(arrange(mutate(isochrone_ghs_aoi,area =
st_area(isochrone_ghs_aoi$geom)),desc(area))$ds_cnes[1])` with the highest lack of
redundancy reached an area that was `r round(arrange(mutate(isochrone_ghs_aoi,area =
st_area(isochrone_ghs_aoi$geom)),desc(area))$area[1]/arrange(mutate(isochrone_ghs_aoi,
area = st_area(isochrone_ghs_aoi$geom)),area)$area[1],2)` times larger than that of the
hospital `r arrange(mutate(isochrone_ghs_aoi,area =
st_area(isochrone_ghs_aoi$geom)),area)$ds_cnes[1]`.
```

2. Quarto code chunk parameters

235c255

```
< #| eval: false
```

```
---
```

```
> #| eval: true
```

3. Renaming figures directories

125c145,146

```
< knitr::include_graphics("figures/area_study_v7.png")
```

```
---
```

```
> library(knitr)
```

```
> knitr::include_graphics("~/agile-gscience-2024-rs-flood/code/figures/area_study_v7.png")
```

155c176

```
< knitr::include_graphics("figures/research_design_v6.png")
```

```
---
```

```
>
```

```
knitr::include_graphics("~/agile-gscience-2024-rs-flood/code/figures/research_design_v6.png")
```

195c216

```
< knitr::include_graphics("figures/weight_sampling_osm_v2.png")
```

```
---
```

```
>
```

```
knitr::include_graphics("~/agile-gscience-2024-rs-flood/code/figures/weight_sampling_osm_v2.png")
```

224c245

```
< knitr::include_graphics("figures/snapping_point_isochrone_v5.png")
```

```
---
```

```
>
```

```
knitr::include_graphics("~/agile-gscience-2024-rs-flood/code/figures/snapping_point_isochrone_v5.png")
```

255c276

```
< knitr::include_graphics("figures/rq1_gc_flow_results_v7.png")
```

```
---
```

```
>
knitr::include_graphics("~/agile-gscience-2024-rs-flood/code/figures/rq1_gc_flow_results_v7.
png")
301c322
< knitr::include_graphics("figures/pre_gc_map_v7.png")
---
> knitr::include_graphics("~/agile-gscience-2024-rs-flood/code/figures/pre_gc_map_v7.png")
309c330
< knitr::include_graphics("figures/rq3_v6.png")
---
> knitr::include_graphics("~/agile-gscience-2024-rs-flood/code/figures/rq3_v6.png")
```

4. Minor errors

```
245c266
< After the flood, the centrality values of the remaining `r
round(units::drop_units(sum(st_length(filter(centrality_pre, !(Id %in%
st_intersection(centrality_pre[centrality_pre$Id %in% filter(centrality_pre, centrality >=
natural_jenks_breaks_cmc_pre[4])$Id,], flood_extent)$Id) & centrality >=
natural_jenks_breaks_cmc_pre[4])$geom)))/1000)` km of urban arteries fell below `r
natural_jenks_breaks_cmc_pre[4]` and were distributed over lower intervals reducing their
total CMC by `r round((sum(centrality_post[centrality_post$Id %in% (filter(centrality_pre, !(Id
%in% st_intersection(centrality_pre[centrality_pre$Id %in% filter(centrality_pre, centrality >=
natural_jenks_breaks_cmc_pre[4])$Id,], flood_extent)$Id) & centrality >=
natural_jenks_breaks_cmc_pre[4])$Id,]$centrality)- sum(filter(centrality_pre, centrality >=
natural_jenks_breaks_cmc_pre[4])$centrality)) / sum(filter(centrality_pre, centrality >=
natural_jenks_breaks_cmc_pre[4])$centrality),2)`%.
---
> After the flood, the centrality values of the remaining `r
round(units::drop_units(sum(st_length(filter(centrality_pre, !(Id %in%
st_intersection(centrality_pre[centrality_pre$Id %in% filter(centrality_pre, centrality >=
natural_jenks_breaks_cmc_pre[4])$Id,], flood_extent)$Id) & centrality >=
natural_jenks_breaks_cmc_pre[4])$geom)))/1000)` km of urban arteries fell below `r
natural_jenks_breaks_cmc_pre[4]` and were distributed over lower intervals reducing their
total CMC by `r round((sum(centrality_post[centrality_post$Id %in% (filter(centrality_pre, !(Id
%in% st_intersection(centrality_pre[centrality_pre$Id %in% filter(centrality_pre, centrality >=
natural_jenks_breaks_cmc_pre[4])$Id,], flood_extent)$Id) & centrality >=
natural_jenks_breaks_cmc_pre[4])$Id,]$centrality)- sum(filter(centrality_pre, centrality >=
natural_jenks_breaks_cmc_pre[4])$centrality)) / sum(filter(centrality_pre, centrality >=
natural_jenks_breaks_cmc_pre[4])$centrality),2)*100`%.
```

```
248c269
< The low connectivity roads, with CMC between `r natural_jenks_breaks_cmc_pre[1]+1`
and `r natural_jenks_breaks_cmc_pre[2]`, reduced their length by `r
units::drop_units(round(sum(st_length(filter(centrality_pre, centrality <=
natural_jenks_breaks_cmc_pre[2])$geom))/1000)) -
units::drop_units(round(sum(st_length(centrality_post[centrality_post$Id %in%
filter(centrality_pre, centrality <= natural_jenks_breaks_cmc_pre[2])$Id,]$geom))/1000))` km
```

```
(`r round((units::drop_units(sum(st_length(centrality_post[centrality_post$Id %in%
filter(centrality_pre, centrality <=
natural_jenks_breaks_cmc_pre[2])$Id,]$geom))) - units::drop_units(sum(st_length(filter(centra
lity_pre, centrality <= natural_jenks_breaks_cmc_pre[2])$geom)))) /
units::drop_units(sum(st_length(filter(centrality_pre, centrality <=
natural_jenks_breaks_cmc_pre[2])$geom))), 2) * 100`%). However, their core metropolitan
connectivity increased by `r round((sum(filter(df_centrality_both_sankey, centrality_pre <=
natural_jenks_breaks_cmc_pre[2] & centrality_post != 0)$centrality_post) -
sum(filter(df_centrality_both_sankey, centrality_pre <= natural_jenks_breaks_cmc_pre[2] &
centrality_post != 0)$centrality_pre)) / sum(filter(df_centrality_both_sankey, centrality_pre <=
natural_jenks_breaks_cmc_pre[2] & centrality_post != 0)$centrality_pre) * 100, 2)`%, driven
by `r units::drop_units(round(sum(st_length(filter(centrality_post, (Id %in%
filter(centrality_pre, centrality >= natural_jenks_breaks_cmc_pre[2])$Id) & centrality <=
natural_jenks_breaks_cmc_pre[2])$geom)) / 1000)) +
units::drop_units(round(sum(st_length(filter(centrality_post, !(Id %in% centrality_pre$Id) &
centrality <= natural_jenks_breaks_cmc_pre[2])$geom)) / 1000))` km of intermediate and
high-value and unused roads moving into this interval. Part of the `r
units::drop_units(round(sum(st_length(filter(centrality_post, !(Id %in%
centrality_pre$Id))$geom)) / 1000))` km of previously unused roads with 0 core metropolitan
connectivity values before the flood became part of roads with higher centralities values.
```

```
> The low connectivity roads, with CMC between `r natural_jenks_breaks_cmc_pre[1]+1`
and `r natural_jenks_breaks_cmc_pre[2]`, reduced their length by `r
units::drop_units(round(sum(st_length(filter(centrality_pre, centrality <=
natural_jenks_breaks_cmc_pre[2])$geom)) / 1000)) -
units::drop_units(round(sum(st_length(centrality_post[centrality_post$Id %in%
filter(centrality_pre, centrality <= natural_jenks_breaks_cmc_pre[2])$Id,]$geom)) / 1000))` km
(`r round((units::drop_units(sum(st_length(centrality_post[centrality_post$Id %in%
filter(centrality_pre, centrality <=
natural_jenks_breaks_cmc_pre[2])$Id,]$geom))) - units::drop_units(sum(st_length(filter(centra
lity_pre, centrality <= natural_jenks_breaks_cmc_pre[2])$geom)))) /
units::drop_units(sum(st_length(filter(centrality_pre, centrality <=
natural_jenks_breaks_cmc_pre[2])$geom))), 2) * 100`%). However, their core metropolitan
connectivity increased by `r round((sum(filter(df_centrality_both_sankey, centrality_pre <=
natural_jenks_breaks_cmc_pre[2] & centrality_post != 0)$centrality_post) -
sum(filter(df_centrality_both_sankey, centrality_pre <= natural_jenks_breaks_cmc_pre[2] &
centrality_post != 0)$centrality_pre)) / sum(filter(df_centrality_both_sankey, centrality_pre <=
natural_jenks_breaks_cmc_pre[2] & centrality_post != 0)$centrality_pre) * 100)`%, driven by
`r units::drop_units(round(sum(st_length(filter(centrality_post, (Id %in% filter(centrality_pre,
centrality >= natural_jenks_breaks_cmc_pre[2])$Id) & centrality <=
natural_jenks_breaks_cmc_pre[2])$geom)) / 1000)) +
units::drop_units(round(sum(st_length(filter(centrality_post, !(Id %in% centrality_pre$Id) &
centrality <= natural_jenks_breaks_cmc_pre[2])$geom)) / 1000))` km of intermediate and
high-value and unused roads moving into this interval. Part of the `r
units::drop_units(round(sum(st_length(filter(centrality_post, !(Id %in%
centrality_pre$Id))$geom)) / 1000))` km of previously unused roads with 0 core metropolitan
connectivity values before the flood became part of roads with higher centralities values.
```


261c282

< Given the strategic function of these arterial roads for urban mobility in the metropolitan region of Porto Alegre, the flood reduced the core metropolitan connectivity of these roads by a ``r (sum(centrality_post[centrality_post$Id %in% (filter(centrality_pre, !(Id %in% st_intersection(centrality_pre[centrality_pre$Id %in% filter(centrality_pre, centrality >= natural_jenks_breaks_cmc_pre[4])$Id,], flood_extent)$Id) & centrality >= natural_jenks_breaks_cmc_pre[4])$Id,]$centrality)- sum(filter(centrality_pre, centrality >= natural_jenks_breaks_cmc_pre[4])$centrality)) / sum(filter(centrality_pre, centrality >= natural_jenks_breaks_cmc_pre[4])$centrality))`%`.

> Given the strategic function of these arterial roads for urban mobility in the metropolitan region of Porto Alegre, the flood reduced the core metropolitan connectivity of these roads by a ``r round((sum(centrality_post[centrality_post$Id %in% (filter(centrality_pre, !(Id %in% st_intersection(centrality_pre[centrality_pre$Id %in% filter(centrality_pre, centrality >= natural_jenks_breaks_cmc_pre[4])$Id,], flood_extent)$Id) & centrality >= natural_jenks_breaks_cmc_pre[4])$Id,]$centrality)- sum(filter(centrality_pre, centrality >= natural_jenks_breaks_cmc_pre[4])$centrality)) / sum(filter(centrality_pre, centrality >= natural_jenks_breaks_cmc_pre[4])$centrality),2)`%`.

294c315

< In contrast, municipalities with low population density per km², such as Viamão with ``r filter(connectivity_municipality_lc_rq, municipality=="Viamão")$density`` inhabitants per km², were correlated with higher intracity connectivity in the post-disaster scenario.

> In contrast, municipalities with low population density per km², such as Viamão with ``r round(filter(connectivity_municipality_lc_rq, municipality=="Viamão")$density)`` inhabitants per km², were correlated with higher intracity connectivity in the post-disaster scenario.

324c345

< While the ``r length(resilience_visintini_table[st_contains(st_buffer(st_transform(centro_historico_porto,31982), 3200), st_transform(resilience_visintini_table,31982), sparse= FALSE),]$ds_cnes)`` hospitals within 3.2 km from the ``r centro_historico_porto$nm_bairro`` of Porto Alegre scored an average of ``r round(mean(resilience_visintini_table[st_contains(st_buffer(st_transform(centro_historico_porto,31982), 3200), st_transform(resilience_visintini_table,31982), sparse= FALSE),]$lack_pre))`` in the lack of redundancy, the remaining ``r length(resilience_visintini_table[!st_contains(st_buffer(st_transform(centro_historico_porto,31982), 3200), st_transform(resilience_visintini_table,31982), sparse= FALSE),]$ds_cnes)`` healthcare facilities scored an average of ``r round(mean(resilience_visintini_table[!st_contains(st_buffer(st_transform(centro_historico_porto,31982), 3200), st_transform(resilience_visintini_table,31982), sparse= FALSE),]$lack_pre))`` reaching the highest lack of redundancy at the hospital ``r string::str_to_title(resilience_visintini_table[which(st_distance(centro_historico_porto, resilience_visintini_table)==max(st_distance(centro_historico_porto, resilience_visintini_table))),]$ds_cnes)``, ``r`

units::drop_units(round(max(st_distance(centro_historico_porto, resilience_visintini_table))/1000))` km away from the Centro Histórico of Porto Alegre.

```
> While the `r
length(resilience_visintini_table$ds_cnes[st_contains(st_buffer(st_transform(centro_historico_porto,31982), 3200), st_transform(resilience_visintini_table,31982), sparse= FALSE)])`
hospitals within 3.2 km from the `r centro_historico_porto$nm_bairro` of Porto Alegre scored
an average of `r
round(mean(resilience_visintini_table$lack_pre[st_contains(st_buffer(st_transform(centro_historico_porto,31982), 3200), st_transform(resilience_visintini_table,31982), sparse=
FALSE)]))` in the lack of redundancy, the remaining `r
length(resilience_visintini_table$ds_cnes[!st_contains(st_buffer(st_transform(centro_historico_porto,31982), 3200), st_transform(resilience_visintini_table,31982), sparse= FALSE)])`
healthcare facilities scored an average of `r
round(mean(resilience_visintini_table$lack_pre[!st_contains(st_buffer(st_transform(centro_historico_porto,31982), 3200), st_transform(resilience_visintini_table,31982), sparse=
FALSE)]))` reaching the highest lack of redundancy at the hospital `r
stringr::str_to_title(resilience_visintini_table[which(st_distance(centro_historico_porto,
resilience_visintini_table)==max(st_distance(centro_historico_porto,
resilience_visintini_table))),]$ds_cnes)` , `r
units::drop_units(round(max(st_distance(centro_historico_porto,
resilience_visintini_table))/1000))` km away from the Centro Histórico of Porto Alegre.
```

237c257,258

<

```
> library(tidyverse)
> library(BAMMtools)
```

331c352

```
< The hospitals most affected by the flood were the `r
stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto Alegre" & lack_post
== 0)$code[1])` `r stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto
Alegre" & lack_post == 0)$ds_cnes[1])` and the `r
stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto Alegre" & lack_post
== 0)$code[2])` `r stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto
Alegre" & lack_post == 0)$ds_cnes[2])`, with `r
stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto Alegre" & lack_post
== 0)$beds[1])` and `r stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto
Alegre" & lack_post == 0)$beds[2])` patients respectively.
```

```
> The hospitals most affected by the flood were the `r
stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto Alegre" & lack_post
== 0)$code[1])` `r stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto
Alegre" & lack_post == 0)$ds_cnes[1])` and the `r
stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto Alegre" & lack_post
== 0)$code[2])` `r stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto
Alegre" & lack_post == 0)$ds_cnes[2])`, with `r
```



```
stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto Alegre" & lack_post == 0)$beds[1])` and `r stringr::str_to_title(filter(resilience_visintini_table, shapename=="Porto Alegre" & lack_post == 0)$beds[2])` beds respectively.
```

5. Loading temp data

94a95,114

```
> ``{r}
> # Load all the temp_data ending in gpkg saving you time
> library(tictoc)
> library(tidyverse)
> library(sf)
> tic()
> temp_data <- list.files(path=
"/home/ricardors/agile-gscience-2024-rs-flood/data/temp_data", pattern="\\.gpkg")
> for (i in seq_along(temp_data)) {
>   file_name <- tools::file_path_sans_ext(temp_data[i])
>   assign(file_name, sf::read_sf(paste0("~/agile-gscience-2024-rs-flood/data/temp_data/",
temp_data[i])))
> }
> # Load non-gpkg files
> centrality_both_sankey_v3 <-
read.csv("~/agile-gscience-2024-rs-flood/data/temp_data/centrality_both_sankey_v3.csv")
> df_centrality_both_sankey <-
read.csv("~/agile-gscience-2024-rs-flood/data/temp_data/df_centrality_both_sankey.csv")
> # Work on flat
> sf::sf_use_s2(FALSE)
> toc()
> ``
>
```

6. Calculating results

259c280

< In terms of core metropolitan connectivity (CMC), the flooding also blocked urban arteries that had a high edge betweenness centrality value, such as Avenida Presidente Castelo Branco and BR-290 Freeway (shown in Fig 6, resulting in a severe **59% loss** of centrality.

> In terms of core metropolitan connectivity (CMC), the flooding also blocked urban arteries that had a high edge betweenness centrality value, such as Avenida Presidente Castelo Branco and BR-290 Freeway (shown in Fig 6, resulting in a severe **r**
 $\text{round}((\text{sum}(\text{centrality_post}\$centrality) - \text{sum}(\text{centrality_pre}\$centrality)) / \text{sum}(\text{centrality_pre}\$centrality), 2) * 100$ % loss of centrality.

354c375

< Unexpectedly, **r** $\text{round}(\text{units::drop_units}(\text{sum}(\text{st_length}(\text{filter}(\text{centrality_post}, !(\text{Id \%in\% centrality_pre}\$Id) \& \text{centrality} \geq \text{natural_jenks_breaks_cmc_pre}[2])\$geom)) / 1000))$ km (**r**

```
round(round(units::drop_units(sum(st_length(filter(centrality_post, !(Id %in%
centrality_pre$Id) & centrality > natural_jenks_breaks_cmc_pre[2])$geom))/1000))/
round(units::drop_units(sum(st_length(filter(centrality_post, !(Id %in%
centrality_pre$Id))$geom))/1000))*100)`%) of the `r
round(units::drop_units(sum(st_length(filter(centrality_post, !(Id %in%
centrality_pre$Id))$geom))/1000))` km of unused roads had relatively high centrality values
(>726) for the post disaster IC average.
```

```
> Unexpectedly, `r round(units::drop_units(sum(st_length(filter(centrality_post, !(Id %in%
centrality_pre$Id) & centrality >= natural_jenks_breaks_cmc_pre[2])$geom))/1000))` km (`r
round(round(units::drop_units(sum(st_length(filter(centrality_post, !(Id %in%
centrality_pre$Id) & centrality > natural_jenks_breaks_cmc_pre[2])$geom))/1000))/
round(units::drop_units(sum(st_length(filter(centrality_post, !(Id %in%
centrality_pre$Id))$geom))/1000))*100)`%) of the `r
round(units::drop_units(sum(st_length(filter(centrality_post, !(Id %in%
centrality_pre$Id))$geom))/1000))` km of unused roads had relatively high centrality values
(`r natural_jenks_breaks_cmc_pre[2]`) for the post disaster IC average.
```